Bi-Directional Synchronization

Bi-directional synchronization maintains data consistency
between the logical SysML v2 architecture and the physical 3D CAD model.

- **Structural Instantiation**: Generate CAD assembly hierarchies derived from SysML v2 block structures.
- **Parameter Propagation**: Update the system model with (computed) mechanical properties from the CAD environment (material, size).
- **Constraint Verification**: Evaluate CAD-derived parameters against defined system boundaries.
- **Impact Traceability**: Correlate requirement modifications with affected geometric entities.

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Bi-Directional Synchronization

1. Automated Target–Actual Verification

Values are pulled from the CAD model and checked automatically against a limit or agreement defined in the system model.

- **Budget tracking (ongoing development)**
  - Mass, installation space, ... checked daily against subsystem allocations
  - Replaces manually maintained spreadsheets with lead times of several weeks

    →Model based, not document based
  - Benefit: early warning: overruns surface immediately
  - Requires: limits as machine-checkable rules

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Bi-Directional Synchronization

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- **Interface control (distributed development)**
  - Compares agreed interface dimensions against as-designed geometry across subsystem boundaries
  - Surfaces unilateral dimension changes without coordination
  - Effective when multiple design owners work in parallel

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- **Replacement part approval (in-service phase)**
  - Verifies whether a substitute material or alternative component still meets the original requirements
    - Triggered by end-of-life notices, supply chain changes or forced material substitution
  - Verifies whether the replacement still meets the original requirements and system limits

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- **Substance compliance (continuous)**
  - environmental protection regulations thresholds and recycling ratios stored as requirements and checked automatically
  - Underlying material and mass data already exists in the CAD model

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Bi-Directional Synchronization

2. Change Impact Analysis

The link between requirement and component is evaluated to derive impact instead of estimating it.

- **Component → requirement:** identifies which requirements and verification cases need re-assessment
- *Example:* switching a housing from aluminium to plastic

  → looks like a local design change, but affects mass, thermal path and material certification

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Bi-Directional Synchronization

3. Evidence and Compliance Reporting

Verification results from the first two groups are captured directly as review and certification evidence, replacing manual compilation of documentation.

- Design state X verified against requirement state Y, with this result
- Reduces compilation effort for design reviews and, in regulated domains, for certification

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Bi-Directional Synchronization
Feasibility

- https://www.youtube.com/watch?app=desktop&v=f906vNYMDJY&ra=m#bottom-sheet
- MCP-Demo of Dassault system with CAD and SysML-V2 Connections
- Can update and read parameters in both systems via one prompt
  - ➢ So synchronization should also be possible

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Bi-Directional Synchronization

System Architecture

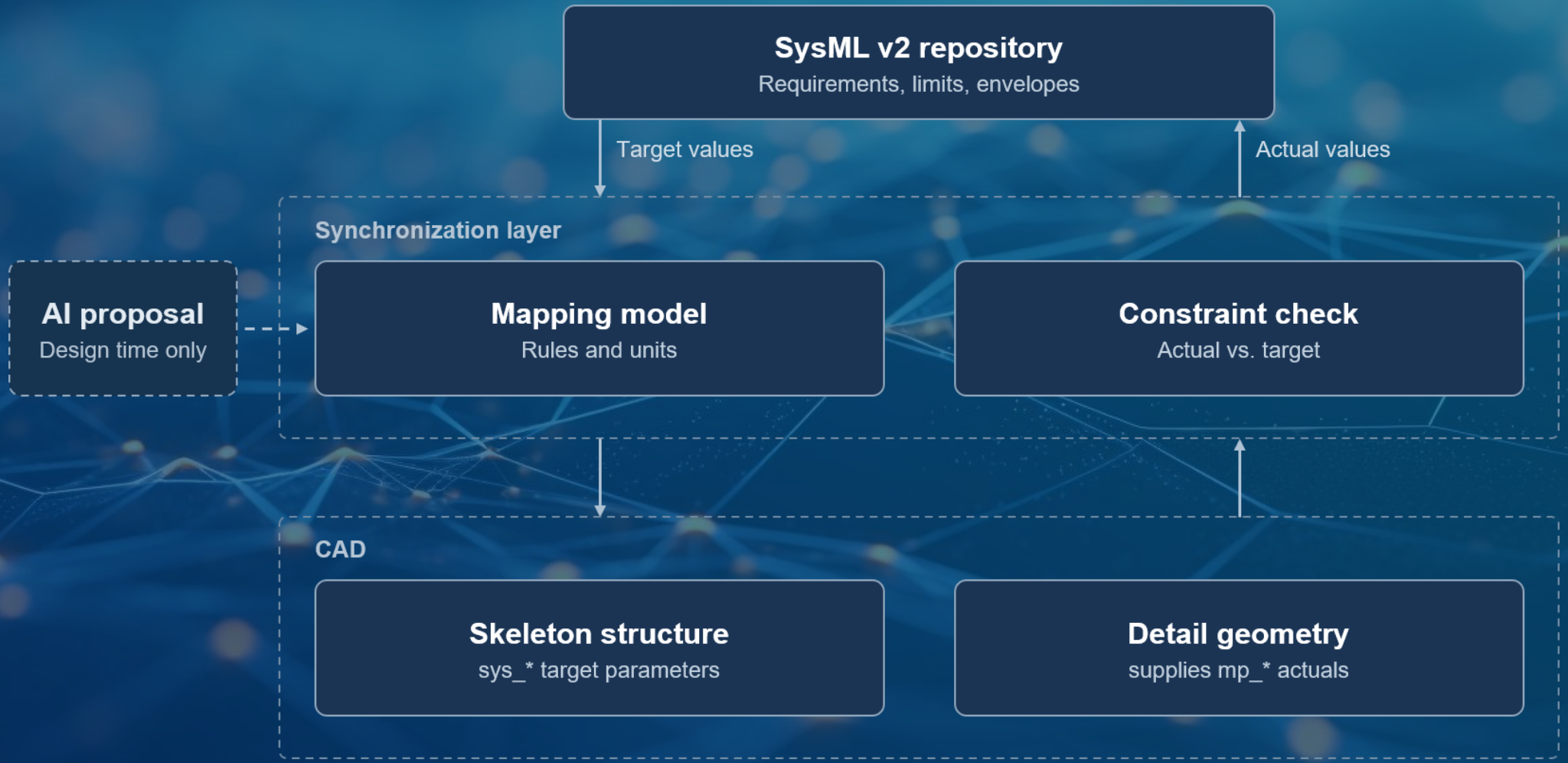
- AI can read both systems and then do the Automated Target–Actual Verification use cases
- Through reading and comparison can also write reports
- For change impact analysis AI can find mappings between CAD and SysML-Models
  - ➢ Instantiate concrete Adapter per Attribute that then makes the syncs deterministic for auditability

SysML-V2	CAD
part def	Component definition
interface/connect	Assembly constraint
Attribute	Parameter

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Bi-Directional Synchronization

System Architecture



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